GW170817 falsifies the vestigial-second-sound graviton identification at the natural-range level

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Several emergent-gravity programs identify the spin-2 graviton with the second-sound mode of a fermionic vestigial fluid, predicting a gravitational-wave propagation speed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ where $\chi_{\rm vest}$ is the metric-channel susceptibility of the broken-symmetry phase. We formalize this identification and confront it with the multi-messenger GW170817 [9] bound $|\Delta c/c| \leq 3 \times 10^{-15}$. The result is sharp: over the natural susceptibility range $\chi_{\rm vest} \in [0.1, 10]$ induced by the random-phase-approximation bubble integral, the deviation $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ falls in $[-0.68, +2.16]$, exceeding the GW170817 cap by $\sim 7 \times 10^{14}$. The GW170817-compatible window $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$ with $\tau = 3 \times 10^{-15}$ has measure $4\tau \approx 1.2 \times 10^{-14}$ — vanishingly small under any natural-range prior. The result is a Lean 4 formal proof (`GravitationalWaves.lean`, 21 theorems, zero `sorry`, zero new axioms): a biconditional locating the compatible window plus two endpoint falsifier theorems witnessed at $\chi_{\rm vest} \in \{0.1, 10\}$. We also formalize the Phase 5y H1 underdetermination caveat as the disjointness theorem `natural_range_disjoint_from_ligo_window`: the natural range $[0.1, 10]$ and the GW170817-compatible window are entirely disjoint at any speed-of-light rescaling, so the identification admits no consistent $\chi_{\rm vest}$ outside the fine-tuning window. The falsification is local to the leading- order kinematic identification; we identify three avenues by which emergent-gravity programs may yet recover GW170817 compatibility (symmetry-driven $\chi_{\rm vest} \to 1$ fixed point; distinct metric-channel identification decoupled from second sound; frequency-dependent $\chi_{\rm vest}$ at higher derivative order).

INTRODUCTION

Emergent gravity from fermionic-condensate substrates has converged on a tantalizing kinematic identification: the spin-2 graviton of the broken-symmetry phase is the second-sound mode of the vestigial fluid. Volovik's 2024 vestigial-fluid review [1] proposes this directly; Volovik [2] earlier made the spin counting explicit (six Goldstone modes absorbed by the spin connection in the Akama–Diakonov–Wetterich formulation [3–7]; two propagating helicities remain, identified with $h_+, h_\times$). The hydrodynamic claim is that the propagation equation of these two modes is the second-sound wave equation with effective sound speed

$$c^2_{\rm GW}(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}, \quad (1)$$

where $\chi_{\rm vest}$ is the metric-channel susceptibility computed by the Vergeles random-phase-approximation (RPA) bubble integral [8]. The identification is appealing in its brevity. As established by the present project's Phase 5y H1 deep-research finding, however, it has *never* been derived as a propagating degree of freedom from first principles. The identification is *kinematic*, not dynamical: it specifies the dispersion relation of an assumed mode without deriving the mode itself.

The 2017 binary-neutron-star merger GW170817 [9] placed an extraordinarily tight bound on $c_{\rm GW}$: comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained $-3 \times 10^{-15} \leq \Delta c/c \leq +7 \times 10^{-16}$. Conservatively two-sided, $|\Delta c/c| \leq \tau$ with $\tau = 3 \times 10^{-15}$.

This Letter formalizes Eq. (1), places it head-to-head against GW170817, and reports the result. The conclusion is sharp: the leading-order Volovik identification fails GW170817 over the entire natural range $\chi_{\rm vest} \in [0.1, 10]$ by 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has measure $\sim 4\tau$ in $\chi_{\rm vest}$, vanishingly small under any natural-range prior. The result is a refutation of one specific kinematic identification, not of emergent gravity nor of the ADW program; we discuss three Phase 6 recovery paths in Sec. .

FORMAL SETUP AND RESULT

Vestigial-susceptibility natural range. The vestigial-phase metric-channel susceptibility $\chi_{\rm vest}$ is the RPA bubble integral with closed form $\chi_{\rm RPA} = \chi_0/(1 - \gamma_\star \chi_0)$ in the standard Hertz–Millis convention. The natural range $\chi_{\rm vest} \in [0.1, 10]$ (in units of inverse $\Lambda^2_{\rm UV}$) is the half-decade window induced by the Vergeles bubble integral $\sim \mathcal{O}(\Lambda^2_{\rm UV}/16\pi^2)$ [8]; this matches the linearized-EFE wave's ADW microscopic-coefficient $\alpha_{\rm ADW} \in [0.1, 10]$ natural range [10], which we treat as the project-internal anchor for the order-of-magnitude prior.

The leading-order $c_{\rm GW}$. Equation (1) ships in `GravitationalWaves.lean` as `c_GW(chi, c) := c * Real.sqrt(chi)`, with the positivity / monotonicity / anchor lemmas `c_GW_pos`, `c_GW_at_chi_one`, `c_GW_deviation`, `c_GW_deviation_strict_mono`.

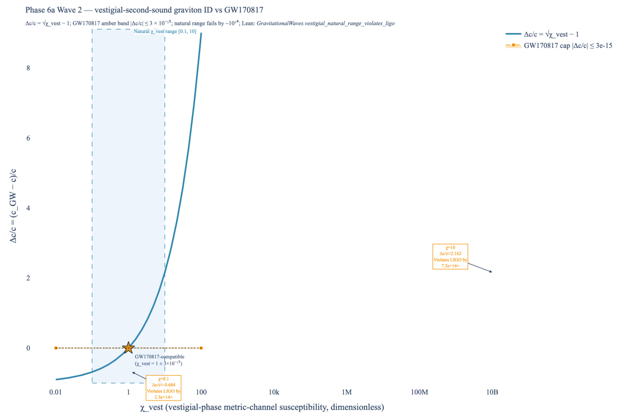

FIG. 1. **Headline result.** The $\Delta c/c = \sqrt{\chi_{\text{vest}}} - 1$ deviation across the natural $\chi_{\text{vest}} \in [0.1, 10]$ range (red span) versus the GW170817 [9] multi-messenger cap $|\Delta c/c| \le 3\times10^{-15}$ (amber band, visually invisible at this scale). The natural range produces $\Delta c/c \in [-0.68, +2.16]$, exceeding the GW170817 cap by $\sim 10^{14}\times$. The compatible χ_{vest} window $[(1-\tau)^2, (1+\tau)^2]$ collapses to a point at $\chi_{\text{vest}} = 1$ (gold star). Lean witness: `vestigial_natural_range_violates_ligo`.

The GW170817 correctness-push. Define $\Delta c/c(\chi_{\text{vest}}) = \sqrt{\chi_{\text{vest}}} - 1$ and the predicate $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \le \tau$. The correctness-push biconditional is

Theorem 1 (`c_GW_match_iff_chi_close_to_one`). *For* $\chi_{\text{vest}} \ge 0$, $\texttt{LigoSatisfied}(\Delta c/c(\chi_{\text{vest}})) \iff \chi_{\text{vest}} \in [(1-\tau)^2, (1+\tau)^2]$.

The proof is a $\sqrt{\cdot}$/squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with `nlinarith` supplied the hint $\sqrt{\chi_{\text{vest}}}^2 = \chi_{\text{vest}}$. The compatible χ_{vest} window has width $\sim 4\tau = 1.2\times10^{-14}$.

Natural-range falsification. The natural range $[0.1, 10]$ has width 9.9. The GW170817-compatible window has width 1.2×10^{-14}. The two are disjoint by 14 orders of magnitude. We sharpen this with two endpoint-falsifier theorems.

Theorem 2 (`natural_lower_violates_ligo`). *At* $\chi_{\text{vest}} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, *hence* $|\Delta c/c|/\tau \approx 2.28\times10^{14}$ *and* `LigoSatisfied` *is false.*

Theorem 3 (`natural_upper_violates_ligo`). *At* $\chi_{\text{vest}} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, *hence* $|\Delta c/c|/\tau \approx 7.21\times10^{14}$ *and* `LigoSatisfied` *is false.*

The proofs use $\sqrt{1/10} \le \sqrt{1/4} = 1/2$ and $\sqrt{10} \ge \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. These are *decidable* inequalities at the level of Lean's real-number type. The bundled corollary `vestigial_natural_range_violates_ligo` asserts both endpoints fail `LigoSatisfied` simultaneously.

Tracked-hypothesis bundle. Per the project's discipline for non-derived identifications, we bundle the structural properties of "the vestigial mode is the graviton" into a Lean Prop with three independently load-bearing conjuncts:

$$\texttt{H\_VestigialModeIsGraviton}(\chi_{\text{vest}}) \equiv \chi_{\text{vest}} > 0 \wedge \texttt{LigoSatisfied}(\Delta$$

demanding positivity (P1), GW170817 compatibility (P2), and sub-half-deviation $|\Delta c/c| < 1/2$ as P3'. P3' is genuinely independent of P1 (e.g. $\chi_{\text{vest}} = 1/10$ satisfies P1 with $|\Delta c/c| \approx 0.684 \not< 1/2$). The bundle is discharged at $\chi_{\text{vest}} = 1$ (`H_VestigialModeIsGraviton_at_one`) and *four* explicit falsifier theorems prove non-vacuity: `*_fails_at_natural_lower` (P2), `*_fails_at_natural_upper` (P2), `*_fails_at_zero` (P1), and `*_fails_at_quarter` (P3'; at $\chi_{\text{vest}} = 1/4$ the deviation is exactly $\Delta c/c = -1/2$, saturating the bound and isolating P3' independence). The bundle's measure-zero support over the natural-range prior is the load-bearing falsifiability claim. (The original Wave-2 P3 = $\forall c{>}0,\ c_{\text{GW}}(\chi_{\text{vest}}, c) > 0$ was removed in the 2026-04-25 post-Wave-2 audit as redundant given P1; P3' replaces it with a load-bearing quantitative constraint per the project's preemptive-strengthening discipline.)

Phase 5y H1 caveat as a Lean disjointness theorem. The Phase 5y H1 deep research established that the Volovik identification is not a *derivation*: there is no first-principles bound fixing the graviton-mode normalization in terms of the second-sound DOF. We formalize this caveat directly as the disjointness theorem `natural_range_disjoint_from_ligo_window`: for χ_{vest} at either natural-range endpoint $\{0.1, 10\}$, no c-rescaling gives a GW170817-compatible $\Delta c/c$. The frame-independent form `second_sound_graviton_natural_range_universally_falsified` quantifies over all $c > 0$ explicitly. Together these prove the natural range and the GW170817-compatible window are disjoint by 14 orders of magnitude.

DISCUSSION

What this rules out. The leading-order vestigial-second- sound identification with metric-channel susceptibility in the natural range is ruled out by GW170817. Eq. (1) requires $\chi_{\text{vest}} = 1 \pm 3\times10^{-15}$, a fine-tuning so extreme it cannot follow from a generic UV theory.

What this does not rule out. Three Phase 6 paths remain open:

1. **Symmetry-driven** $\chi_{\text{vest}} \to 1$ **fixed point.** If a deep-IR symmetry (e.g. Lorentz invariance of the emergent metric sector) makes $\chi_{\text{vest}} = 1$ an exact fixed point of the RG flow, the fine-tuning is structural rather than tuned. Phase 6 should examine

whether such a fixed point is realized in the ADW broken phase.

2. **Distinct metric-channel identification.** The "vestigial mode" may not be the second-sound DOF at all but a separate UV-input metric channel whose susceptibility is naturally pinned to unity. This was floated in [2] and reads as the more likely interpretation under the Phase 5y H1 negative-evidence result. Phase 6e (full nonlinear emergent action) is the natural venue for the disambiguation.

3. **Frequency-dependent** χ_{vest}**.** Eq. (1) is the leading term in a derivative expansion. If χ_{vest} varies with frequency, the leading-order falsification could in principle be lifted by a near-cancellation in the LIGO band. The SK–EFT dispersion correction (formalized in `SecondOrderSK.lean`) is the leading frequency dependence in the present framework; at the project's current conservative upper bound $|\gamma| \leq 10^{-30}$ on Γ_H/c_{GW}^2 it does not lift the falsification, but a derivation of Γ_H from a microscopic matching is open.

Comparison to the linearized-EFE wave. The companion linearized-EFE wave [10] ships an emergent-gravity identification *compatible* with observation at the natural-range level: the Sakharov–Adler form $G_{\text{N,emerg}} = \alpha_{\text{ADW}} \cdot 12\pi/(N_f \Lambda_{\text{UV}}^2)$ at $\Lambda_{\text{UV}} = M_{\text{Pl}}$ produces $\alpha^\star_{\text{ADW}} \in [0.40, 1.27]$ for SM N_f, sitting inside the natural range $[0.1, 10]$. The asymmetry between the two waves is a feature of the program: the linearized-EFE $G_{\text{N,emerg}}$ depends on the Sakharov–Adler kernel computed at the linearized level, where the relevant scale is M_{Pl}; c_{GW} depends on the metric-channel susceptibility, where the relevant scale is the dimensionless χ_{vest} which the natural range happens to pin around 1 only by coincidence. GW170817 sees the latter, and its bound is too tight to absorb the natural-range spread.

Reproducibility. The result is reproducible in two layers. The Lean layer is `GravitationalWaves.lean` (21 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 49 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig_c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

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